

Duplicate.java

main()

array [10, 20, 40, 80]

for(int n=10; n<=200; n*=2)

1. $n=10$

① $10 < 200 \checkmark$ $n = 10 \times 2$

$n=20$

② $20 < 200 \checkmark$ $n = 20 \times 2$

$n=40$

③ $40 < 200 \checkmark$ $n = 40 \times 2$

$n=80$

④ $80 < 200 \checkmark$ $n = 80 \times 2$

$n=160$

⑤ $160 < 200 \checkmark$ $n = 160 \times 2$

$n=320$

$320 < 200 \times$

loop run 5 times

• distinct use all elements are unique.

hasDuplicateNested()

$n=10$

$i=0$ $j \rightarrow 1, 2, 3, 4, 5, 6, 7, 8, 9$

9 compar

$i=1$ $j \rightarrow 2, 3, 4, 5, 6, 7, 8, 9$

⑧

$i=2$ $j \rightarrow 3, 4, 5, 6, 7, 8, 9$

⑦

⑥

!

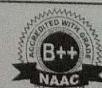
;

$i=8$ $j=9$

①

$i=9$ no value.

ADSUL'S TECHNICAL CAMPUS



$$50 \quad 9+8+7+6+5+4+3+2+1$$

$$45$$

n	nested loop comparison
10	45
20	190
40	780

$$\frac{n(n-1)}{2} \quad \text{constant and dominant will take}$$

$$O(n^2)$$

hasDuplicateSet()

suppose

a [10, 20, 30, 40]

x=10 — seen.add(10) ✓ continue
 x=20 — seen.add(20) ✓
 x=30 — seen.add(30) ✓
 40 — ✓

every value is unique.

!seen.add(x) ✗ never true

so it return false

n	HashSet operation
10	10
20	20
30	30

$O(n)$

